IxLoad Automation

Solution Architect - Automation
Hubert Gee



AGENDA

- Automating IxLoad in Windows and Linux Gateway
- API browser
- Sample script walkthrough
- Sample scripts from Github/OpenIxia

IxLoad Automation Platforms

MULTIPLE CHOICES TO FIT YOUR ENVIRONMENT

Run scripts from:

- Windows
- Linux

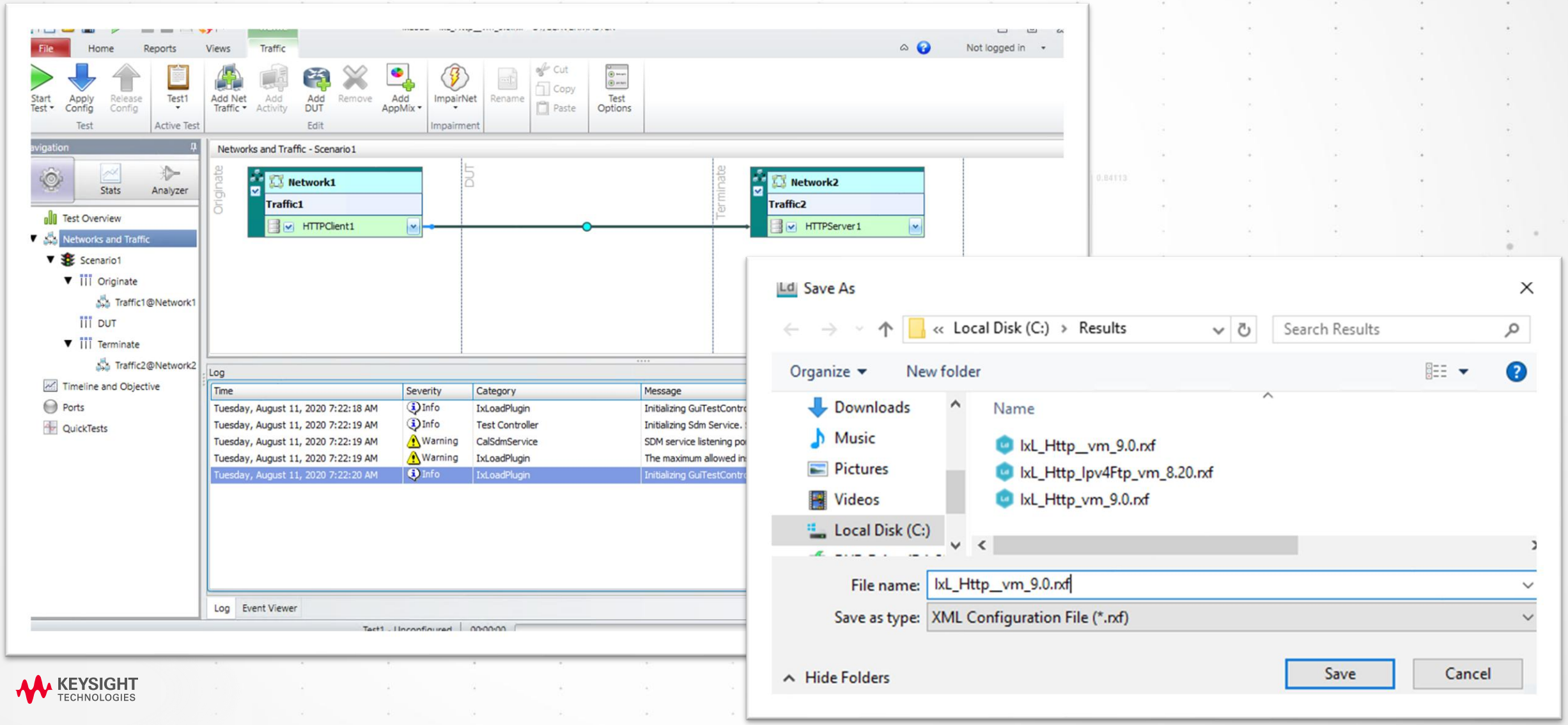
ReST API Gateway Options	Platform & Benefits
IxLoad / Windows Gateway	• Windows OS • API browser • 10+ sessions
IxLoad / Linux Gateway	• OVA and QCOW2 • API browser • 10+ sessions

Chassis Options

- Windows Chassis
- Linux Chassis
- Virtual Chassis
  - KVM
  - vmware

Use IxLoad Windows GUI to Create The Configuration

SAVE THE CONFIG TO A FILE



KEYSIGHT TECHNOLOGIES

Script

Use the sample script: LoadConfigFile.py

https://github.com/OpenIxia/IxLoad/tree/master/RestApi/Python/SampleScripts/LoadSavedConfigFile

- LoadConfigFileEvalStats.py
  - Allows defining stat expectations

IxLoad Linux Gateway For Automation

API BROWSER IN WEB UI

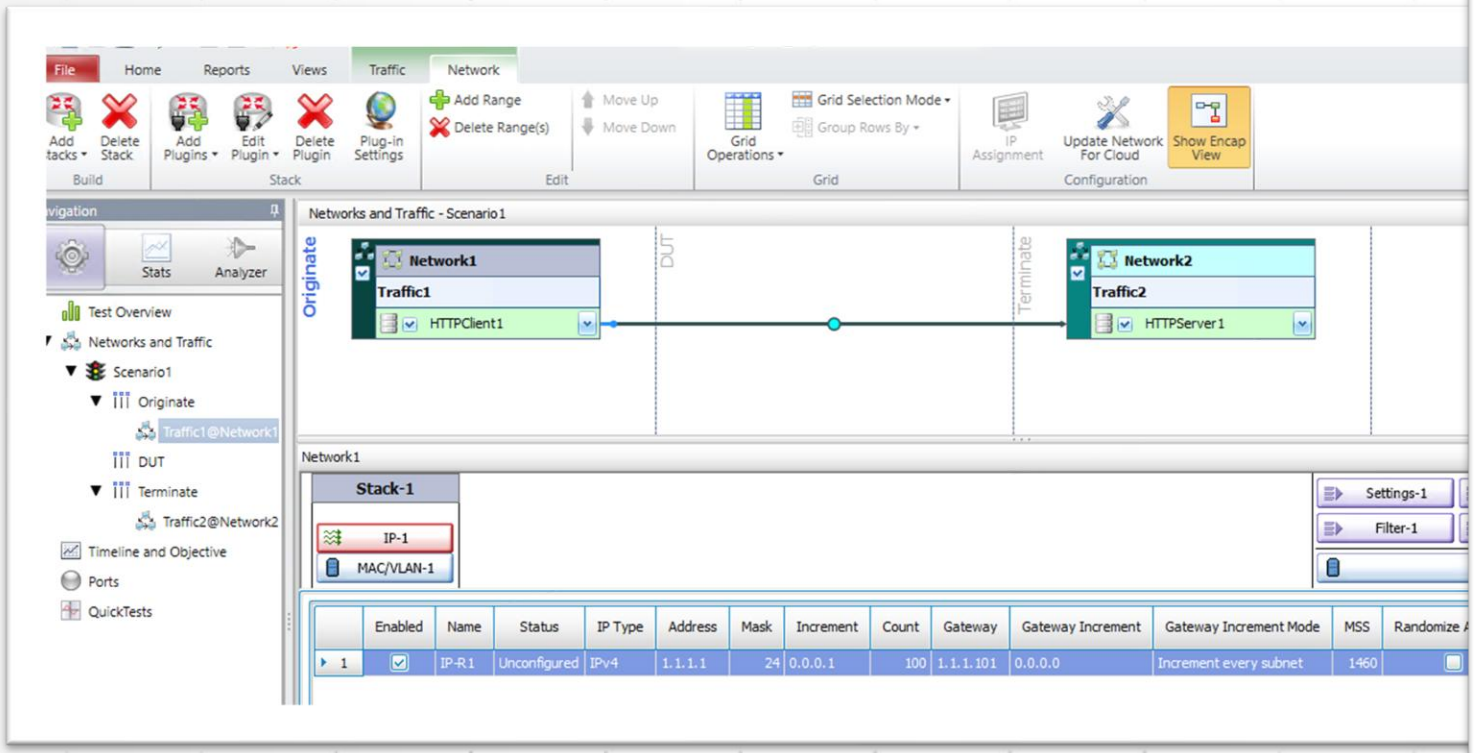
- Load a configuration:
  - Run test and view stats
  - Or use the API browser to view configurations for APIs

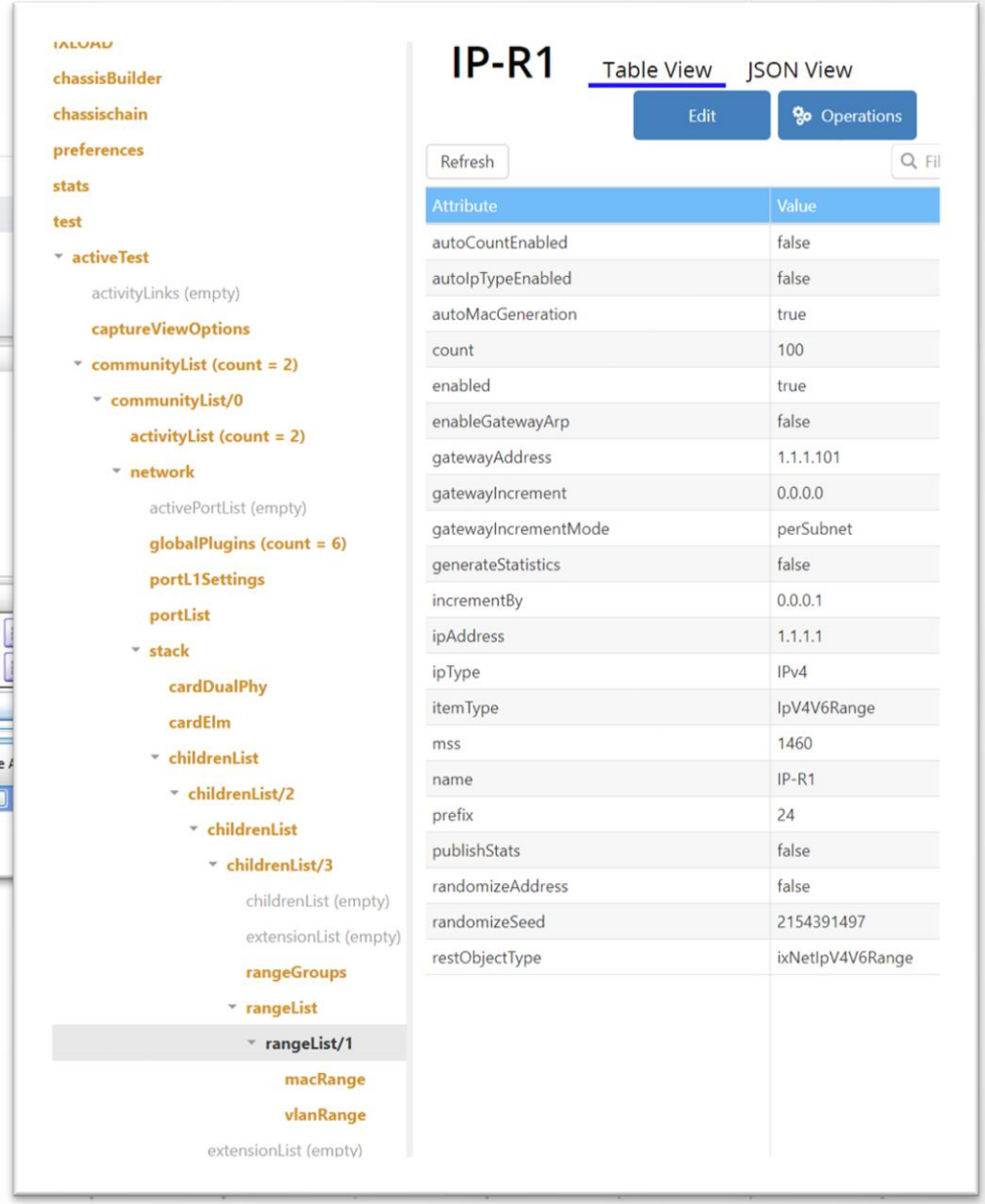
- Supports 10+ simultaneous sessions for automation

- Secured access: Use HTTPS (8443) Example: https://192.168.70.2:8443
- Non secured access: Use HTTP (8080) Example: http://192.168.70.2:8080

API Browser demo

IxLoad GUI and API Browser Correlation

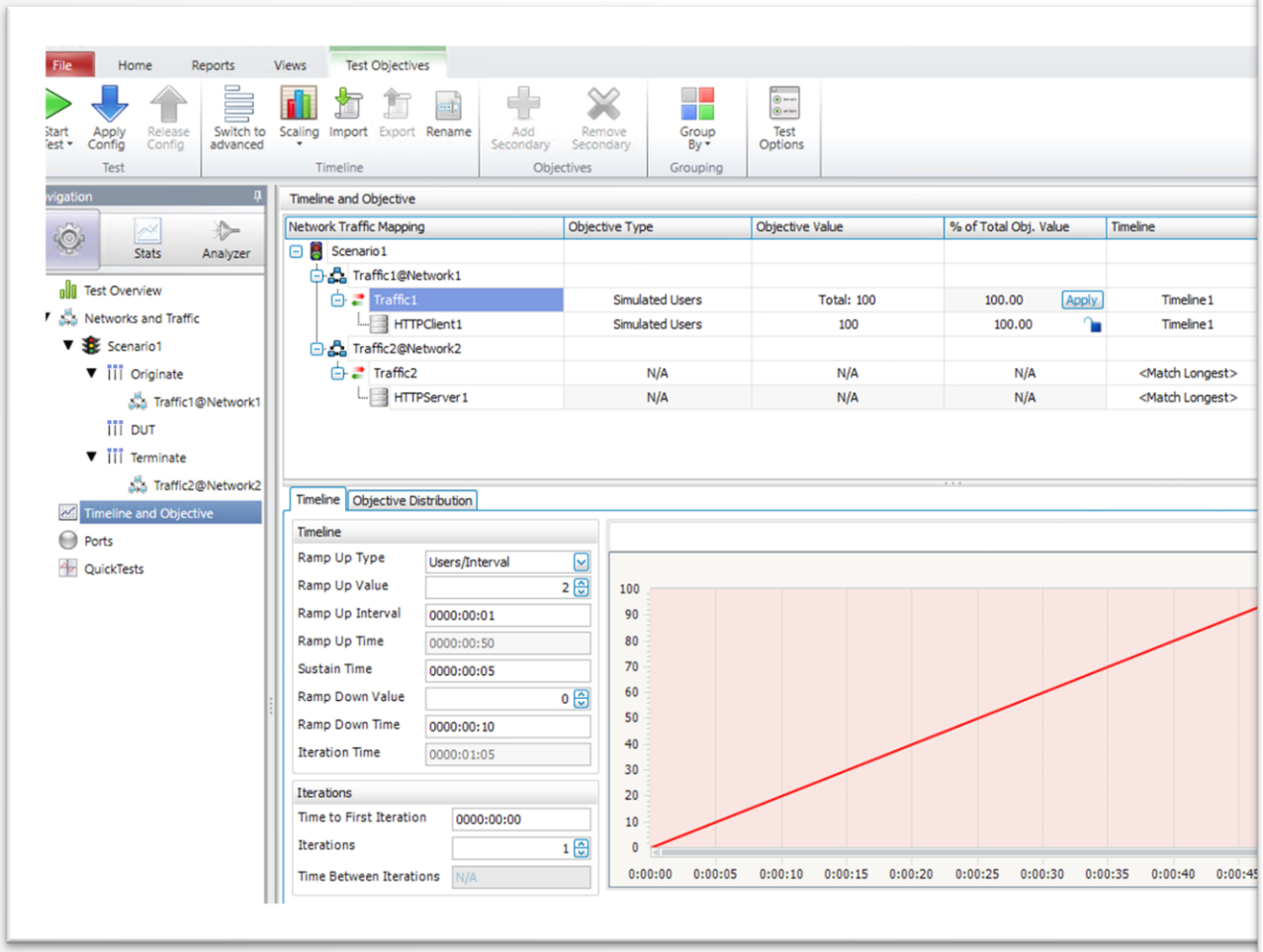


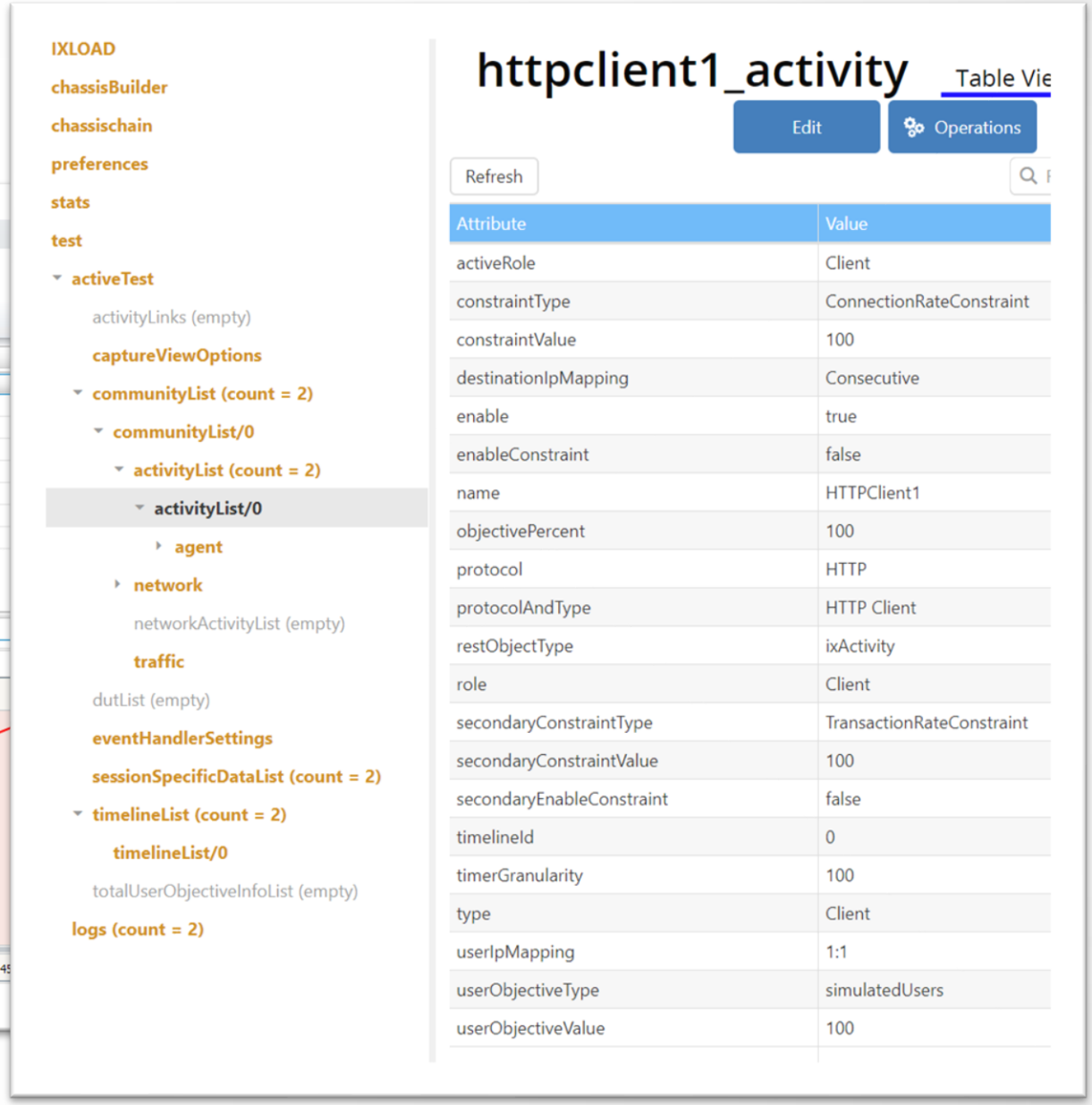


KEYSIGHT TECHNOLOGIES

IxLoad GUI and API Browser Correlation

OBJECTIVE/CONSTRAINT

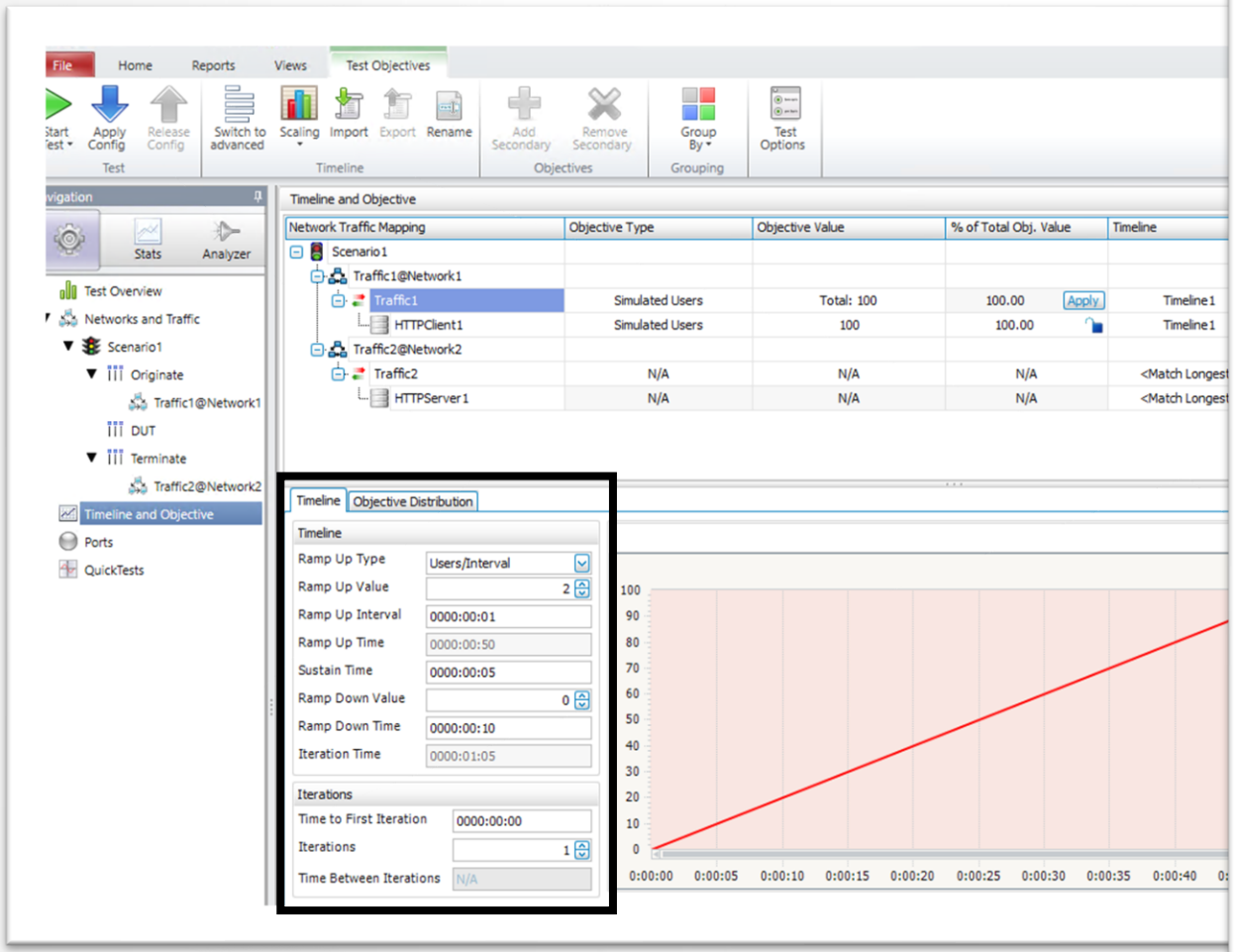


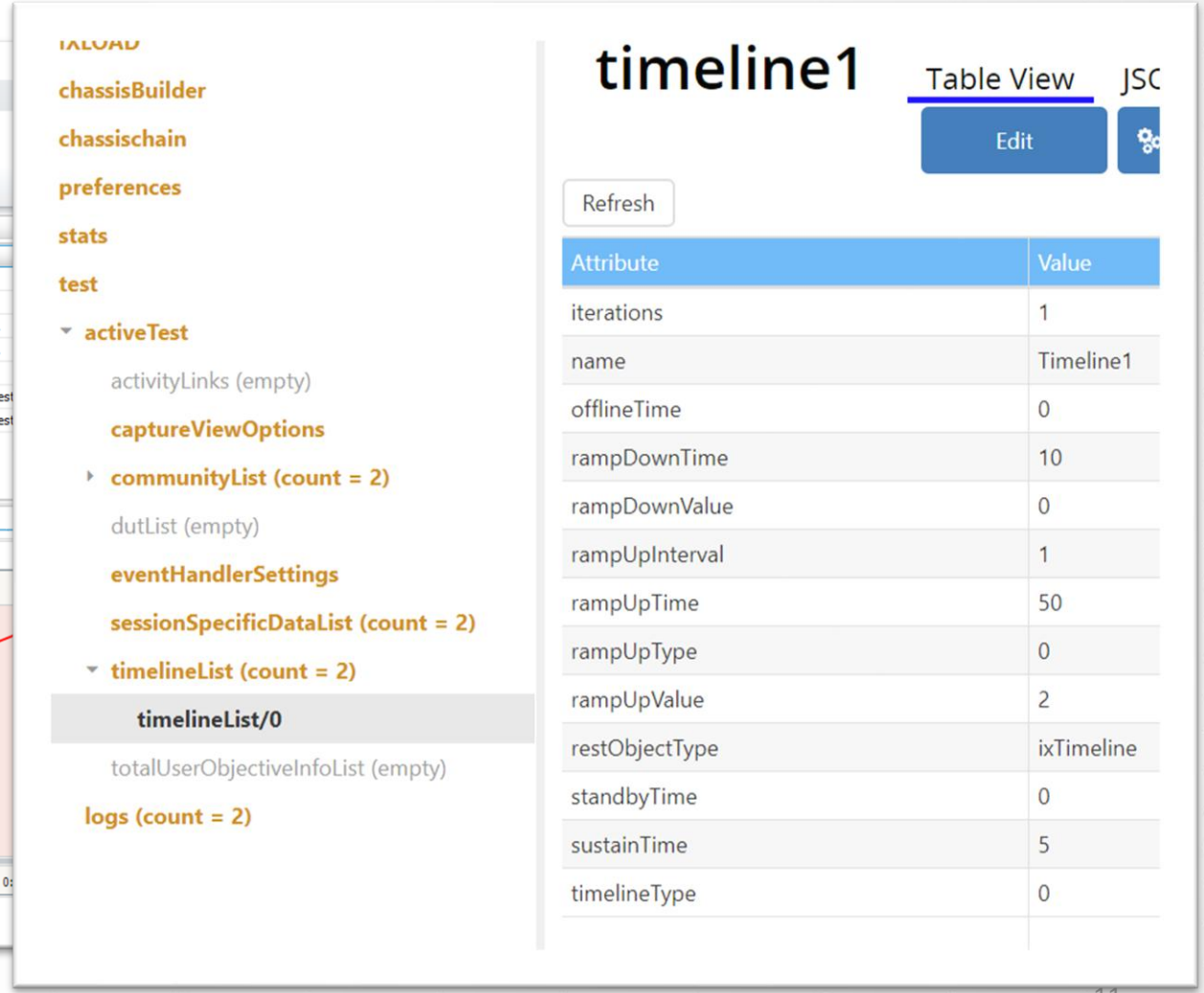


KEYSIGHT TECHNOLOGIES

IxLoad GUI and API Browser Correlation

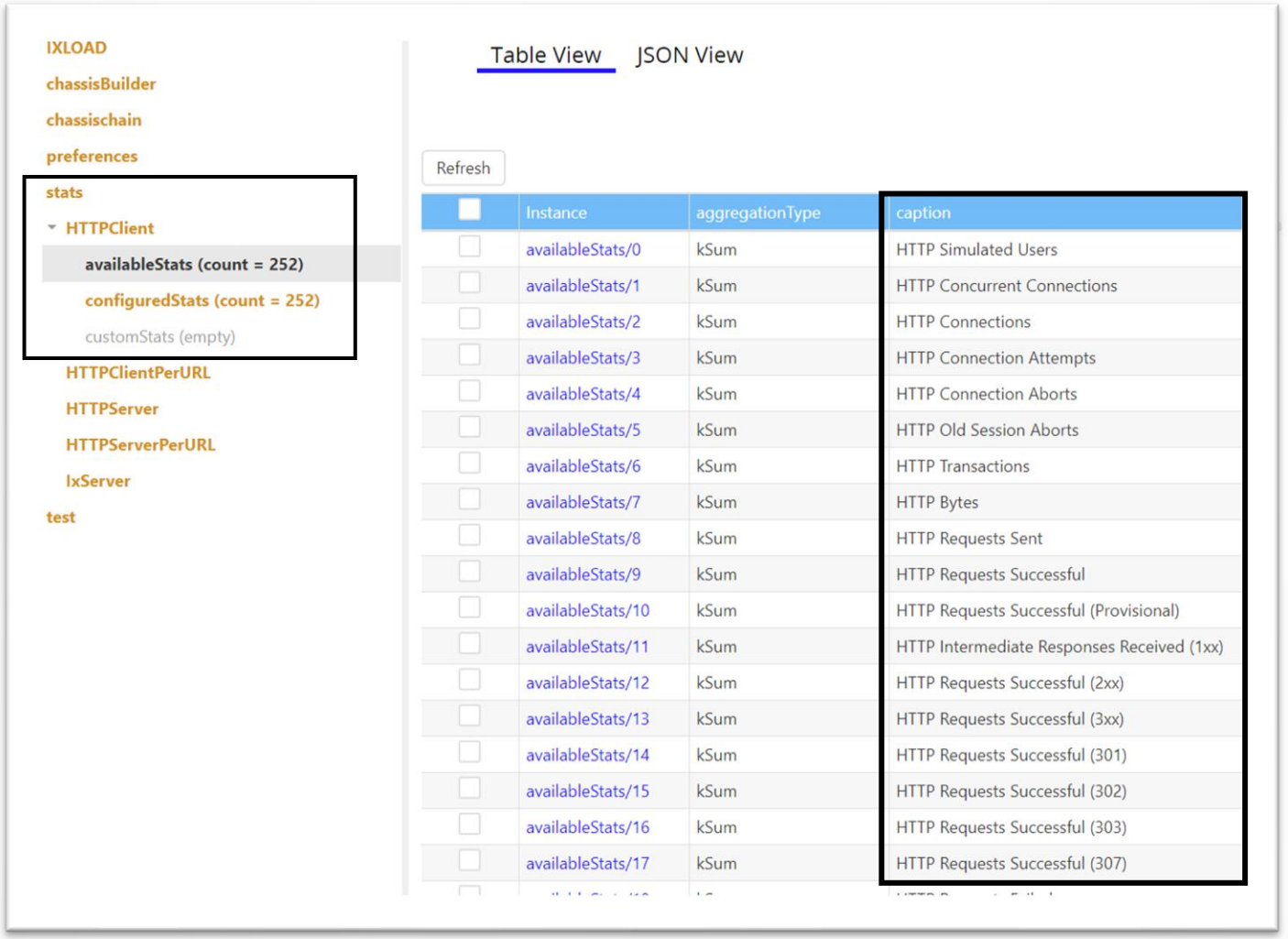
TIMELINE





IxLoad API Browser

STATISTIC NAMES



Instance	aggregationType	caption
availableStats/0	kSum	HTTP Simulated Users
availableStats/1	kSum	HTTP Concurrent Connections
availableStats/2	kSum	HTTP Connections
availableStats/3	kSum	HTTP Connection Attempts
availableStats/4	kSum	HTTP Connection Aborts
availableStats/5	kSum	HTTP Old Session Aborts
availableStats/6	kSum	HTTP Transactions
availableStats/7	kSum	HTTP Bytes
availableStats/8	kSum	HTTP Requests Sent
availableStats/9	kSum	HTTP Requests Successful
availableStats/10	kSum	HTTP Requests Successful (Provisional)
availableStats/11	kSum	HTTP Intermediate Responses Received (1xx)
availableStats/12	kSum	HTTP Requests Successful (2xx)
availableStats/13	kSum	HTTP Requests Successful (3xx)
availableStats/14	kSum	HTTP Requests Successful (301)
availableStats/15	kSum	HTTP Requests Successful (302)
availableStats/16	kSum	HTTP Requests Successful (303)
availableStats/17	kSum	HTTP Requests Successful (307)

Automating IxLoad

Use the sample script: LoadConfigFile.py

https://github.com/OpenIxia/IxLoad/tree/master/RestApi/Python/SampleScripts/LoadSavedConfigFile

- Create a configuration in Windows IxLoad GUI.
- Save the configuration to a .rxf file
- For VoIP, save the configuration to a .crf file

- Get familiar with the script. Connect two back-2-back Ixia ports.
- Use the script **LoadConfigFile.py** to load a saved config file.
- Assign chassis and ports
- <Make necessary configuration modifications>
- Start test
- Poll real time stats and evaluate stats
- Retrieve CSV statistic results from the IxLoad gateway server

Note: The saved config version is compatible with future version releases, but it is not compatible in prior releases. For example. A config file saved in 9.0 will not work on IxLoad 8.50.

Set Statistics For Run Time Evaluation

```
# Get run time stat results with an operator and the expected value for the stat
# operators:  None, >, <, =, !=, <=, >=
statsDict = {
    'HTTPClient': [{'caption': 'TCP Connections Established', 'operator': '>', 'expect': 60},
                   {'caption': 'HTTP Simulated Users',        'operator': '>=', 'expect': 100},
                   {'caption': 'HTTP Connections',            'operator': '>', 'expect': 300},
                   {'caption': 'HTTP Transactions',           'operator': '>', 'expect': 190},
                   {'caption': 'HTTP Connection Attempts',    'operator': '>', 'expect': 300}
                  ],
    'HTTPServer': [{'caption': 'TCP Connections Established',    'operator': '>', 'expect': 1000},
                   {'caption': 'TCP Connection Requests Failed', 'operator': '=', 'expect': 0}
                  ]
```

Polling Stats and Evaluating Stats At Run Time

```
HTTPClient:
        TCP Connections Established: 1728
                Passed: Expecting: >60

        HTTP Simulated Users: 0
                Failed: Expecting: >=100

        HTTP Connections: 1728
                Passed: Expecting: >300

        HTTP Transactions: 1728
                Passed: Expecting: >190

        HTTP Connection Attempts: 1728
                Passed: Expecting: >300


HTTPServer:
        TCP Connections Established: 1728
                Passed: Expecting: >1000

        TCP Connection Requests Failed: 0
                Passed: Expecting: =0
```

Final Report

```
HTTPClient
        Failed: TCP Connections Established
        Failed: HTTP Simulated Users
        Failed: HTTP Connections
        Failed: HTTP Transactions
        Failed: HTTP Connection Attempts

HTTPServer
        Failed: TCP Connections Established
        Passed: TCP Connection Requests Failed
```

Script walkthrough

Leverage the IxL_RestApi.py Module

USING PYTHON INHERITANCE

Create a new module. For example: my_ixload_module.py

This library module is from Ixia
import IxL_RestApi

Create your own class and inherit Ixia's library module as the base module
Class CustomAPI(**IxL_RestApi.Main**)

```
    def __init__(self, *args, **kwargs):
        super().__init__(*args, **kwargs)
```

Continue …

Leverage the IxL_RestApi.py Module

USING PYTHON INHERITANCE

```
# Create a new module. For example:  my_ixload_module.py

import IxL_RestApi

Class CustomAPI(IxL_RestApi.Main)

    def __init__(self, *args, **kwargs):
        super().__init__(*args, **kwargs)

    # Create your own custom functions
    def  custom_function(self):

        #    Your function's logic goes here
        #    Use the base IxL_RestApi.Main() functions to execute your IxLoad ReST APIs using:  get(), post(), patch() and delete().
        #    Use self.logInfo(), self.logError() to capture print statements into the log file
```

Continue …

Leverage the IxL_RestApi.py Module

USING PYTHON INHERITANCE

Create a new module. For example: myIxLoadApiModuly.py

```
import IxL_RestApi

Class CustomAPI(IxL_RestApi.Main)

    def __init__(self, *args, **kwargs):
        super().__init__(*args, **kwargs)s

    def  custom_function(self):

        #    Your function's logic goes here
        #    Use the base IxL_RestApi.Main() functions to execute your IxLoad ReST APIs using:  get(), post(), patch() and delete().
        #    Use self.logInfo(), self.logError() to capture print statements into the log file

    def pollStats(self):
        # You could also overwrite any function in the Main class. This pollStats is an example
        # to overwrite the pollStats function in IxL_RestApi.py.
        # Do this if there is a bug in IxL_RestApi or if you need to enhance it to fit your requirements.
```

Continue …

Leverage the IxL_RestApi.py Module

BY USING PYTHON INHERITANCE

In a script

From my_ixload_module import *

Before

```
restObj = Main(apiServerIp=apiServerIp,
               apiServerIpPort=apiServerIpPort,
               osPlatform=serverOs,
               deleteSession=deleteSession,
               generateRestLogFile=True)
```

Now

```
restObj = CustomAPI(apiServerIp=apiServerIp,
                    apiServerIpPort=apiServerIpPort,
                    osPlatform=serverOs,
                    deleteSession=deleteSession,
                    generateRestLogFile=True)
```

IxLoad Linux Gateway Installation

1. VMware ova
2. KVM qcow2

- After installing it, power up the IxLoad gateway VM.
- Log into IxLoad with password ixia123
- On the terminal, enter: ip address (To get the dhcp IP address)
- Enter the IP address on a web browser.

References

- The sample script below uses ReST API developed in Python
- Supports python 2.7 and 3+
- Git clone: https://github.com/IxLoad.git

Sample script folder in Github

- https://github.com/OpenIxia/IxLoad/tree/master/RestApi/Python/SampleScripts

Sample scripts: **LoadConfigFile.py**:

- https://github.com/OpenIxia/IxLoad/blob/master/RestApi/Python/SampleScripts/LoadSavedConfigFile/LoadConfigFile.py

The Python library module for scripts: IxL_RestApi.py

- https://github.com/OpenIxia/IxLoad/tree/master/RestApi/Python/Modules


KEYSIGHT
TECHNOLOGIES